

GenomeLens Deterministic Benchmark v1

Frozen public-data technical proof of concept for genomic health-economic decision analysis

GenomeLens is a research platform for translating genomic evidence into decision-relevant health economics: what decision a finding could change, what health and cost consequences could follow, how uncertainty should be represented, and when the evidence is insufficient to assign a defensible economic value.

Deterministic Benchmark v1 evaluates that decision layer across **36 public benchmark genomes**: 30 primary public genomes plus 6 reference/parent genomes. The benchmark is designed to test reproducibility, economic routing, arithmetic integrity, report parity, explicit zero/missingness semantics, and public-report integrity. It is **not** a clinical validation, diagnostic-performance study, population-effectiveness estimate, or regulatory claim.

Frozen v1 results

Validation layer	Frozen v1 result
Cohort completion	36 / 36
Engine	4eaf729
Tooling	a9b3d3a
Full clean-tree suite	7,723 passed, 26 skipped, 0 failed
Release gates	9 / 9
Parity failures	0
Zero audit	0 defects across 1,399 displayed zeros
Arithmetic audit	0 discrepancies
Lost-content audit	0 downgraded rows
Real-row assertions	301 evaluated, 0 failures
PDF semantic checks	36 / 36
Forbidden-string scan	0 hits across 144 rendered PDFs
Analyst challenge review	278 / 278 answered; no P0/P1 finding

Visual-gate disposition

The visual gate flagged **49 of 9,482 rendered pages**, all in the full-genome **report.pdf**. The economics PDFs were clean. The flags consisted of a bounding-box false positive plus minor ancestry-label overlap/clipping and were accepted as a **non-material P2 presentation exception**. This is not represented as an unconditional visual pass.

What GenomeLens demonstrates

GenomeLens separates the genomic evidence state from the economic disposition of a finding. A biologically supported result does not automatically receive a dollar value. Depending on the evidence and decision context, the economic result can be modeled, conditional, unresolved, explicitly zero, unquantified, or refused.

The public benchmark demonstrates two distinct analysis modes:

Individual Genomic Economics

one genome -> evidence-qualified finding -> decision/comparator
-> health consequence -> cost/QALY effect -> ICER/NMB/uncertainty

Reproductive & Carrier Economics

Parent A + Parent B -> inheritance compatibility -> offspring
probability -> reproductive decision -> expected health/economic
consequence

Pair-level reproductive value is treated as a distinct estimand and is not silently added to either parent's personal-health value.

Economic outputs

Where the evidence chain supports quantification, GenomeLens can represent:

- incremental healthcare cost;
- QALY impact;
- ICER;
- net monetary benefit (NMB);
- context-specific or if/when-decision value;
- sensitivity and scenario analysis;
- value-of-information concepts;
- reproductive / family-planning economic consequences;
- explicit true-zero, unresolved, unquantified, and refusal states.

A central benchmark rule is **missing is not zero**: a pathway that lacks the technical resolution, donor context, intervention evidence, or separable economic terms required for a defensible estimate is not silently rendered as \$0.

What the benchmark establishes

- reproducible execution across the intended public-genome cohort;
- consistent routing between genomic evidence states and economic dispositions;
- arithmetic reconciliation for recomputable economic outputs;
- explicit handling of true zero, missing context, conditionality, uncertainty, and refusal;
- report-to-structured-result consistency across public economics PDFs;

- pair-level reproductive decision analysis as a separate unit of analysis;
- resistance to unsupported external analyst suggestions entering the deterministic economics layer.

What it does not establish

- diagnostic sensitivity or specificity of whole-genome sequencing;
- population prevalence or representativeness;
- clinical efficacy for an individual patient;
- a single cash value of a genome;
- personalized medical or reproductive advice;
- regulatory clearance or medical-device validation.

Disclosed release limitations

Two validation activities are explicitly recorded as future work rather than represented as completed:

1. Mutation harnesses were not rerun because their source anchors were stale.
2. No separate full manual adversarial-review pass was completed across every final artifact; one Lynch report was read end-to-end.

In addition, 20 economic rows were kept explicitly classified as not independently recomputable from separable delta-QALY / delta-cost terms rather than being back-filled with fabricated components.

Reproducibility record

- **Benchmark:** GenomeLens Deterministic Benchmark v1
- **Engine commit:** 4eaf729
- **Tooling commit:** a9b3d3a
- **Cohort:** 30 primary public genomes + 6 reference/parent genomes
- Frozen benchmark artifacts were archived and integrity-checked locally.
Internal benchmark packages and raw run outputs are not publicly distributed.
- **Personal/private inputs:** excluded from the public benchmark artifact

Public release scope

Benchmark results, methods, limitations, and reproducibility evidence are public. Proprietary implementation details and private inputs are intentionally excluded.

Status: DETERMINISTIC BENCHMARK v1 - FROZEN

Public-data technical proof of concept. Not clinical validation.